

Sample Paper 8 — Answer Key (Level-2 Difficulty Step-Up)

Total: 60 marks (45×1 mark + 5×3 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **D) Cabbage** — Rose, Lily, and Jasmine are flowers; Cabbage is a vegetable.
2. **A) Sky** — A fish lives in water; a bird lives (flies) in the sky.
3. **A) EYPBCL** — The rule is "-2 letters" (wrapping $A \rightarrow Y$): $G \rightarrow E$, $A \rightarrow Y$, $R \rightarrow P$, $D \rightarrow B$, $E \rightarrow C$, $N \rightarrow L$.
4. **C) 77** — Two patterns are interleaved: the odd-position terms go 6, 12, 18, 24 (+6 each); the even-position terms go 95, 89, 83, ? (-6 each) — so the next even term is 77.
5. **A) Aarav, Dev, Chetna, Bina, Esha** — Chetna is at position 3 (middle). Esha is at the right end, position 5. Dev is immediately left of Chetna, position 2. Bina sits between Esha (5) and Chetna (3), so Bina is position 4. The only person left, Aarav, takes position 1.
6. **A) Back at the starting point** — South and North cancel (14 m each way); East and West cancel (10 m each way).
7. **B) Monday** — $27 - 6 = 21$ days later; 21 is exactly 3 weeks, so the day of the week repeats: Monday.
8. **C) 41** — $16 = 4 \times 4$, $25 = 5 \times 5$, $49 = 7 \times 7$ (each a number multiplied by itself — a "square number"); 41 does not fit this pattern.
9. **C) 960** — Each term is multiplied by one more than the last multiplier used ($\times 2$, $\times 4$, $\times 5$, $\times 6$): $160 \times 6 = 960$.
10. **C) 16th** — The 5 new people join ahead of him at the front end, so 5 more people are now closer to the front than Suresh: his position from the front increases by 5: $11 + 5 = 16$ th.
11. **C) 34** — 19, 23, and 41 are prime numbers; $34 (= 2 \times 17)$ is not.
12. **A) OCTMGV** — The rule is "+2 letters": $M \rightarrow O$, $A \rightarrow C$, $R \rightarrow T$, $K \rightarrow M$, $E \rightarrow G$, $T \rightarrow V$.
13. **D) Cone** — Circle, Square, and Triangle are flat (2-D) shapes; a Cone is a solid (3-D) shape.
14. **C) Team C** — The order from lowest to highest is Team D, Team B, Team A, Team C, so Team C scored the highest.
15. **B) 16** — The gaps between terms are 1, 2, 3, 4 — each one more than the last. The next gap is 5: $11 + 5 = 16$.

Section 2 — Mathematical Reasoning

16. **A) 9,13,468** — $6,78,901 + 2,34,567 = 9,13,468$.

17. **A) 4,83,211** — $8,40,000 - 3,56,789 = 4,83,211$.
18. **A) 1,13,945** — $8,765 \times 13 = 1,13,945$.
19. **A) 7,654** — $1,22,464 \div 16 = 7,654$ exactly.
20. **C) 48** — $96 = 2^5 \times 3$, $144 = 2^4 \times 3^2$. HCF = $2^4 \times 3 = 48$.
21. **C) 60** — $20 = 2^2 \times 5$, $30 = 2 \times 3 \times 5$. LCM = $2^2 \times 3 \times 5 = 60$.
22. **C) 11/13** — As decimals: $7/10 = 0.7$, $4/5 = 0.8$, $11/13 \approx 0.846$, $9/11 \approx 0.818$. The greatest is $11/13$.
23. **C) 8 km 450 m** — $4650 \text{ m} + 3800 \text{ m} = 8450 \text{ m} = 8 \text{ km } 450 \text{ m}$.
24. **B) 1,092 sq m** — Area = $38 \times 29 = 1,092 \text{ sq m}$.
25. **C) 12 cm** — A regular dodecagon has 12 equal sides: $144 \div 12 = 12 \text{ cm}$.
26. **C) 25%** — Profit = $2,500 - 2,000 = ₹500$; profit % = $500/2000 \times 100 = 25\%$.
27. **C) 25%** — If principal P doubles, the interest earned equals P itself. $SI = (P \times R \times T) \div 100 \rightarrow P = (P \times R \times 4) \div 100 \rightarrow R = 100 \div 4 = 25\%$.
28. **C) ₹300** — Total parts = $5 + 6 + 10 = 21$; 1 part = $630 \div 21 = 30$; largest share = $10 \times 30 = ₹300$.
29. **A) 5/8** — $0.625 = 625/1000$, which simplifies to $5/8$.
30. **C) 15 kg** — $6400 \text{ g} + 4750 \text{ g} + 3850 \text{ g} = 15,000 \text{ g} = 15 \text{ kg}$.
31. **C) 15,300** — $4.25 \text{ hours} \times 3600 \text{ seconds/hour} = 15,300 \text{ seconds}$.
32. **B) 81 cm** — An equilateral triangle has 3 equal sides: $3 \times 27 = 81 \text{ cm}$.
33. **C) 1,584 sq cm** — Area = $48 \times 33 = 1,584 \text{ sq cm}$.
34. **B) 8** — A cube has 8 vertices (corners).
35. **C) 20** — Number of diagonals = $n(n-3)/2 = 8 \times 5/2 = 20$.

Section 3 — Everyday Mathematics

36. **C) ₹34,000** — $2,72,000 \div 8 = 34,000$.
37. **C) 1,750** — Rate = $3,150 \div 9 = 350 \text{ bottles/hour}$; in 5 hours = $350 \times 5 = 1,750$.
38. **C) 35 min** — $5,250 \div 150 = 35 \text{ minutes}$.
39. **C) 4 h 37 min** — Total time 7:10 a.m. to 12:05 p.m. = 4 h 55 min; subtracting the 18-minute stop gives 4 h 37 min of actual travel.
40. **C) 2 L 250 mL** — $5200 \text{ mL} + 3800 \text{ mL} = 9000 \text{ mL}$; $9000 \div 4 = 2250 \text{ mL per bottle}$.
41. **C) ₹170** — $7 \times ₹26 = ₹182$; $₹182 - ₹12 \text{ discount} = ₹170$.
42. **B) 1,000 L** — $1/7 \times 1,750 = 250 \text{ L used per day}$; over 3 days = 750 L used; $1,750 - 750 = 1,000 \text{ L remaining}$.
43. **C) 64 kg** — $21300 \text{ g} + 18700 \text{ g} + 13500 \text{ g} + 10500 \text{ g} = 64,000 \text{ g} = 64 \text{ kg}$.
44. **C) 72** — $1,944 \div 27 = 72 \text{ rows}$.
45. **C) ₹300** — Sold for $75 \times ₹44 = ₹3,300$; profit = $₹3,300 - ₹3,000 = ₹300$.

Section 4 — Achievers Section (3 marks each)

46. **C) 18** — Working backwards: $(126 + 36) \div 9 = 162 \div 9 = 18$.
47. **C) 39** — Let the first number be x . Then second = $2x$, third = $2x+15$. Sum: $x + 2x + (2x+15) = 210$
 $\rightarrow 5x + 15 = 210 \rightarrow x = 39$.
48. **C) ₹2,000** — After spending $2/5$ on books, $3/5$ of the money remains. After spending $1/4$ of that on stationery, $3/4$ of $3/5 = 9/20$ of the original remains. So $9/20$ of the total = ₹900 $\rightarrow$ total = ₹2,000.
49. **C) 233** — Boys = $(2 \times 113) + 7 = 233$. Check: $113 + 233 = 346$. ✓
50. **C) 68** — Each term is the sum of the previous two: $26 + 42 = 68$.
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Scoring guide (out of 60, Level-2-difficulty paper): 32+ is a very strong result at this difficulty. 22–31 is solid. Below 22 — review *why* each missed question was harder (an extra step, a bigger number, two ideas combined) rather than the raw score.
